Introduction to the Dataset for INL1 – FOA197

You have been given access to a simulated dataset describing a large number of companies. The dataset contains both predictors (different characteristics of the companies) and two target variables. The variables in the dataset have been anonymized. This means that their names do not reveal what they represent.

Your tasks are:

- Prepare the dataset. The dataset is not ready for analysis. Before you begin modeling, you need to examine the dataset yourselves and make relevant decisions about how to prepare it. This may, for example, involve:
    - identifying and handling missing values,
    - identifying implausible or unusual observations,
    - examining whether certain variables need to be transformed,
    - assessing whether any variables should be excluded from the analysis.
- Explore the variables by examining their distributions, scales, and other relevant characteristics.
- Interpret and rename the variables based on what you reasonably believe they could represent.
- Create a business case in which your interpretations of the variables form a coherent and credible scenario.
- Investigate the relationships between the predictors and the target variables.
- Develop two models – one for pattern recognition and one for prediction – that could be used within your business case. You will find further instructions regarding INL1 on Canvas.

Your choices should be justified based on both the dataset and the business case you have created. Therefore, there is not necessarily a single correct interpretation of the variables or a single correct way to prepare the dataset.

Your interpretations do not need to match the variables' original meanings. What matters is that:

- the interpretation is consistent with the variable's data type, values, and distribution,
- the variables together form a logical and coherent business case,
- the target variables represent outcomes that are relevant for the company to understand or predict,
- you can justify your choices.

Description of the Dataset

The dataset is simulated and contains information about a large number of companies. The original meanings of the variables have been anonymized. They are therefore named Predictor1–Predictor9 and TargetVariable1–TargetVariable2.

Your task is to use the descriptions below to assign reasonable meanings and new names to the variables and use them to create a coherent business case.

There is no answer key specifying exactly what the variables must represent. The same statistical variable can often have several reasonable interpretations. What matters is that your interpretation fits the variable's characteristics and that the variables together form a credible business case.

Target Variables

- **TargetVariable1:** A continuous numerical variable that takes only positive values. Most observations are concentrated around a certain value, while very low and very high values are less common. The variable is suitable for something measured as a quantity that cannot be negative. Examples of possible outcomes include sales, revenue, costs, investments, or production.
- **TargetVariable2:** A binary variable that takes only the values 0 and 1. The variable is therefore suitable for an outcome in which something either occurs or does not occur. Examples of possible outcomes include expansion/no expansion, successful/unsuccessful project, customer loss/no customer loss, or investment/no investment.

Other Variables

- **ID:** A unique number for each observation. The variable is used only to identify observations and has no substantive meaning. For example, it may identify a company.
- **Predictor1:** A binary variable that takes only the values 0 and 1. The variable may represent a characteristic or activity that a company either has or does not have. Examples: participation in an activity, use of a particular system, presence of a policy, or implementation of a measure.
- **Predictor2:** A continuous numerical variable with values between 0 and 100. The variable can be interpreted as an index or a standardized measure in which higher values represent a higher level of some characteristic. Examples: competition, customer satisfaction, digitalization, market potential, or organizational maturity.
- **Predictor3:** A numerical variable consisting mainly of integers and normally expected to range from approximately 18 to 75. Most observations are concentrated around 40. The variable is suitable for something measured in whole units with a naturally limited range. Examples: experience, age, number of employees in a particular function, or another measure that could reasonably fall within this range.

- **Predictor4:** A discrete numerical variable that takes only non-negative integer values. Most observations are relatively close to zero, often between 0 and 15, while a smaller number of observations have higher values. The variable is therefore particularly suitable for something that is counted. Examples: number of new customers, orders, complaints, projects, incidents, or completed activities.
- **Predictor5:** A continuous numerical variable containing both positive and negative values and approximately centered around zero. The variable is suitable for a measure where both positive and negative deviations are meaningful. Examples: change compared with the previous period, deviation from budget, difference from an industry average, or an internal index.
- **Predictor6:** A continuous numerical variable normally expected to be positive. Most observations are around 100, while a smaller number have considerably higher values. The distribution is therefore somewhat right-skewed. The variable is suitable for a measure that cannot be negative and where a small number of companies may have considerably higher values than others. Examples: costs, investments, sales, hours worked, or resource use.
- **Predictor7:** A continuous numerical variable normally expected to be positive. Most observations are at relatively low or moderate levels, while a smaller number of companies have considerably higher values. The distribution is therefore right-skewed. The variable may represent something for which low or moderate levels are common but some companies can have very high levels. Examples: time required, waiting time, costs, workload, or resource consumption.
- **Predictor8:** A categorical variable with three categories: CategoryA, CategoryB, CategoryC. The categories have no natural ranking. The variable is therefore suitable for dividing companies into three different groups. Examples: market segment, industry, geographic region, business model, or customer type.
- **Predictor9:** A categorical variable with three categories: CategoryA, CategoryB, CategoryC. Unlike Predictor8, the categories can be interpreted as having a natural order, from lower to higher levels. Examples: low/medium/high risk, low/medium/high complexity, small/medium/large market, or low/medium/high strategic priority.